

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I CH.HARSHITHA(192473031) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

A. Python programming basics are used

B. Pandas and NumPy are used for data handling

C. Matplotlib and Seaborn are used for result visualization

Aim

To develop a Python-based Student Performance Analysis System that analyzes students' marks and attendance, calculates average marks and pass percentage, and displays the results using graphical visualizations.

Problem Statement

Educational institutions often analyze student performance manually, which is time-consuming and prone to errors. There is a need for a system that can automatically process student data, calculate academic performance, determine pass or fail status, and generate visual reports for better decision-making.

Proposed Solution

The proposed Student Performance Analysis System uses Python to automate the analysis of student records. It reads student data from a CSV file, calculates average marks, identifies pass/fail status, computes the overall pass percentage, and presents the results through charts and graphs. This helps teachers quickly understand student performance and identify areas that need improvement.

Tools and Technologies Used

- **Programming Language:** Python
- **Code Editor:** Visual Studio Code (VS Code)
- **Data Handling:** Pandas
- **Numerical Calculations:** NumPy

- **Data Visualization:** Matplotlib
- **Statistical Visualization:** Seaborn
- **Dataset Format:** CSV (student_data.csv)

CODE:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


data = pd.read_csv("student_data.csv")


data["Average"] = data[["Math", "Science", "English"]].mean(axis=1)


data["Result"] = np.where(data["Average"] >= 50, "Pass", "Fail")


pass_percentage = (data["Result"] == "Pass").mean() * 100


print("==== Student Performance Analysis =====")

print(data)


print("\nAverage Marks of Class:", round(data["Average"].mean(), 2))

print("Highest Average:", round(data["Average"].max(), 2))

print("Lowest Average:", round(data["Average"].min(), 2))

print("Pass Percentage:", round(pass_percentage, 2), "%")


plt.figure(figsize=(8,5))

plt.bar(data["Name"], data["Average"])
```

```
plt.title("Average Marks of Students")
```

```
plt.xlabel("Student Name")
```

```
plt.ylabel("Average Marks")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
plt.figure(figsize=(7,5))
```

```
sns.histplot(data["Average"], bins=5, kde=True)
```

```
plt.title("Distribution of Average Marks")
```

```
plt.xlabel("Average Marks")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

```
plt.figure(figsize=(7,5))
```

```
sns.heatmap(
```

```
    data[["Math", "Science", "English", "Attendance"]],
```

```
    annot=True,
```

```
    cmap="YlGnBu"
```

```
)
```

```
plt.title("Student Performance Heatmap")
```

```
plt.show()
```

EXPECTED OUTCOME:

===== Student Performance Analysis =====

	Name	Math	Science	English	Attendance	Average	Result
0	Alice	85	90	88	95	87.666667	Pass
1	Bob	60	65	58	80	61.000000	Pass
2	Charlie	75	70	72	85	72.333333	Pass

3	David	45	50	40	70	45.000000	Fail
4	Emma	95	92	96	98	94.333333	Pass
5	Frank	55	60	57	78	57.333333	Pass
6	Grace	88	84	90	96	87.333333	Pass
7	Henry	68	72	70	82	70.000000	Pass
8	Ivy	91	94	93	99	92.666667	Pass
9	Jack	49	52	48	75	49.666667	Fail

Average Marks of Class: 71.73

Highest Average: 94.33

Lowest Average: 45.0

Pass Percentage: 80.0 %





